Joseph Walton

AI Engineer / LLM Engineer

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Professional Summary

AI Engineer / LLM Engineer with 10+ years across AI, machine learning, and software engineering, including 8+ years delivering production ML systems in government and industry. Strong track record shipping high-impact AI products (LLM, NLP, CV, search, recommendation) with measurable business outcomes. Experienced in end-to-end production ownership: delivery, reliability, observability, evaluation, and responsible AI governance. Trusted communicator with technical and non-technical stakeholders, including cross-government AI working groups.

Technical Skills

- **LLM / GenAI:** Azure OpenAI, Hugging Face, LangChain, LangGraph, RAG, agentic workflows, prompt engineering, LLM evaluations, observability, red teaming
- **ML / MLOps:** PyTorch, TensorFlow, scikit-learn, MLflow, model deployment, CI/CD for ML, experiment tracking, model monitoring
- **Data / Infrastructure:** Python, SQL, PySpark, Polars, NumPy, Pandas, Elasticsearch, vector search, ETL pipelines, Docker, Git
- **Cloud Platforms:** Azure, Google Cloud Platform, AWS

Professional Experience

IBM

Lead Data Scientist
June 2025 - Present (Contract role, central government)

- Rescued failing AI initiatives by implementing delivery standards for LLM products, including evaluations and observability.
- Designed and delivered an LLM-powered data processing assistant that saves ~2 hours per case worker per day (~£2M annualized value).
- Built a Claude Skill to optimize human-in-the-loop data ingestion for LLM eval creation, reducing evaluation workflow effort by ~1 hour per case.
- Rewrote a data matching pipeline from Apache Spark to Polars, reducing runtime from ~2 hours to ~10 minutes and saving ~£10k/month in cloud costs.

Intellectual Property Office

Senior Machine Learning Engineer

November 2021 - April 2025

- Delivered a GenAI roadmap and production applications to improve trademark examination outcomes.
- Built and deployed LLM-powered services using Azure OpenAI, LangChain, and Azure Functions within a service-oriented architecture.
- Developed a RAG assistant to help novice users identify relevant Goods & Services terminology, improving user task completion.
- Led safe production deployment practices for GenAI, including red teaming, evaluation frameworks, and policy-aligned controls.
- Improved relevance of search and recommendation systems by 20% through data-driven refinement and vector search tuning in Elasticsearch (HNSW).
- Implemented MLOps practices to reduce model deployment lead time to <1 day across multiple environments.
- Replaced a Databricks-heavy ETL pipeline with Polars architecture, reducing compute cost by 50% and enabling zero-downtime deployments.
- Represented UK IPO AI delivery approaches with international IP offices and contributed to cross-government AI best-practice working groups.

Tech Stack: Azure OpenAI, Hugging Face, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

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Incopro

Lead Machine Learning Engineer

February 2020 - November 2021

- Led design and implementation of production ML pipelines using MLOps practices, reducing model release cycle to <1 day.
- Delivered multilingual BERT-based NLP models (90+ languages) for IP infringement detection at scale.
- Implemented text-and-image similarity recommendation systems to improve analyst detection workflows.
- Mentored engineers into machine learning roles to strengthen team delivery capability.

Tech Stack: Vertex AI Pipelines, TensorFlow, PySpark, Google Cloud Platform, MLflow

Machine Learning Engineer

November 2018 - February 2020

- Engineered production NLP systems for infringement prediction, increasing enforced infringements by 10% per analyst (~200 analysts).
- Built and deployed image similarity models that helped secure a major enterprise contract contributing ~10% of ARR.

Tech Stack: TensorFlow, FastText, Google Cloud Platform, MLflow

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Office for National Statistics

Machine Learning Engineer

September 2017 - November 2018

- Built ML pipelines linking large-scale government administrative datasets to support national survey and census operations.

- Developed scalable backend components for Census 2021 data collection.

Secondment: Barclays (Data Scientist, September 2018 - October 2018) - Delivered economic modeling work using payments data as part of ONS x Barclays collaboration.

Tech Stack: Python, PySpark, Cloudera

Purple Secure Systems

Software Engineer
August 2014 - October 2016

- Developed full-stack big-data systems for government and defense clients.

Tech Stack: Java, Python, PySpark, AngularJS

Education

University of Bath

Master of Science in Applied Mathematics
2016 - 2017

Bachelor of Science in Physics
2010 - 2013

Honors & Awards

Winner of ONS x Barclays Data Hackathon

Built payment-data-driven methods to produce more granular and timely estimates for GVA (Gross Value Added).